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**FIRST SEMESTER 2026-27**  
**Course Handout (Part II)**

Date: 13/08/2026

In addition to part I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the courses.

**Course No** : BITS F110  
**Course Title** : ENGINEERING GRAPHICS  
**Instructor-in-charge** : GEETA BHATT  
**Practical Lab Instructors** : Venkata Sai  
Kumar

**1. Course Description:**

The course includes fundamentals and techniques of technical drawing and standard practices of the same so that design ideas can be adequately communicated and produced. It introduces students to the theories of projection and the concepts of engineering drawing using the most widely used CAD application software, AutoCAD. Basic AutoCAD 2023 commands will also be introduced. The course covers the following topics: Introduction to AutoCAD basic commands; theory of projections; orthographic projections; isometric projections; projection of points, lines, planes and solids; section of solids; developments of surfaces.

**2. Scope and objective of the course:**

Computerized drafting provides accurate and easily modifiable graphics entities, easy data storage and retrieval facility and enhances creativity.

Upon successful completion of this course, the student will be able to:

- | Read and interpret engineering drawings
- | Identify the three principal projection planes
- | Draw 2-dimensional orthographic projections from given 3-dimensional views
- | Create an isometric drawing using AutoCAD
- | Comprehend orthographic and multiview projection
- | Apply the concept of cutting planes to create the various types of sectional views
- | Become conversant with appropriate use of AutoCAD software for drafting.

**3. Text Book:** D.M. Kulkarni, A.P. Rastogi and A. K. Sarkar., *Engineering Graphics with AutoCAD*, PHI Learning Private Limited, New Delhi.

**4. Reference Book:** N.D. Bhatt, *Engineering Drawing*, Charotar Publishing House, Gujarat.



## 5. Course Plan:

| Module Number                                             | Lecture session/LabSession                                        | Reference     | Learning Outcome                                                                     |
|-----------------------------------------------------------|-------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------|
| 1. Introduction on Engineering Graphics, Intro to AutoCAD | <b>L1</b> Introduction on Engineering Graphics. Basic Commands.   | Ch-1          | Opening of AutoCAD file, Saving, Editing etc., Basic Commands of drawing, editing.   |
|                                                           | Practice Lab Sessions                                             | Ch-2          |                                                                                      |
| 2. Orthographic projections                               | <b>L2-L4</b> Multi view drawing from given pictorial views.       | Ch-3 and Ch-5 | Learn to draw the orthographic projections for given objects (pictorial views)       |
|                                                           | Practice Lab Sessions                                             |               |                                                                                      |
|                                                           | <b>E1 Evaluative Lab Session</b>                                  | Ch-5          |                                                                                      |
| 3. Isometric Drawing                                      | <b>L5-L6</b> Pictorial View from Multiview Drawing                | Ch-6          | Learn to draw the isometric drawing with the given orthographic views.               |
|                                                           | Practice Lab Sessions                                             |               |                                                                                      |
|                                                           | <b>E2 Evaluative Lab Session</b>                                  |               |                                                                                      |
| 4. Projection of Lines                                    | <b>L7</b> Projection of points; lines, true lengths, inclinations | Ch-9          | Learn to draw projections of lines in planar surfaces placed in different quadrants. |
|                                                           | Practice Lab Session                                              |               |                                                                                      |
|                                                           | <b>E3 Evaluative Lab Session</b>                                  |               |                                                                                      |
| 5. Projection of Planes                                   | <b>L8.</b> Projection of Planes                                   | Ch-10         |                                                                                      |
|                                                           | Practice Lab Sessions                                             |               |                                                                                      |
|                                                           | <b>E4 Evaluative Lab Session</b>                                  |               |                                                                                      |
| 6. Projection of Solids                                   | <b>L9</b> Projections of solids                                   | Ch-12         | Learn to draw the projection of solids and the sectional views of solids.            |
|                                                           | Practice Lab Session                                              |               |                                                                                      |
|                                                           | <b>E5 Evaluative Lab Session</b>                                  |               |                                                                                      |
| 7. Sections of Solids                                     | <b>L10 – L11</b> Section planes and sectional view of Solids.     | Ch-13         |                                                                                      |
|                                                           | Practice Lab Sessions                                             |               |                                                                                      |
|                                                           | <b>E6 Evaluative Lab Session</b>                                  |               |                                                                                      |
| 8. Development of surfaces                                | <b>L12</b> Parallel line method                                   | Ch-14         | Learn to develop the external surfaces of the given objects.                         |
|                                                           | <b>L13</b> Radial line method                                     |               |                                                                                      |
|                                                           | Practice Lab Sessions                                             |               |                                                                                      |
|                                                           | <b>E7 Evaluative Lab Session</b>                                  |               |                                                                                      |



#### 6. Evaluation Scheme:

| Evaluation Component              | Weightage | Duration        | Total Marks (200) | Date of Evaluation             |
|-----------------------------------|-----------|-----------------|-------------------|--------------------------------|
| Mid-Semester Exam                 | 25%       | 60 minutes      | 50                | As Announced in the Timetable  |
| Lab Assignments (Best 4 out of 7) | 35%       | Practical Hours | 70                | As per the Lab evaluation plan |
| Comprehensive Exam                | 40%       | 90 minutes      | 80                | As Announced in the Timetable  |

**7.: Chamber Consultation Hour:** Room No. 6020-C, Thursday, 10:00 – 11:00 AM.

**8. Notices:** All the announcements related to the course will be sent via email.

**9. Make-up policy:** No makeup exams will be conducted for lab class assignments. Make-up request for Mid-Semester / Comprehensive examination must be submitted at least one day before the exam date on the AUGSD portal only.

Instructor-in-Charge  
BITS F110